THE COMPLETE BUSINESS KNOWLEDGE ENCYCLOPEDIA

الموسوعة الشاملة للمعرفة في عالم الأعمال

VOLUME 6

Investment, Capital Markets, Commodities and Market Intelligence

الاستثمار وأسواق رأس المال والسلع وقراءة الأسواق

Every Essential Business Concept, Term, Process, Document, Financial Instrument, and Commercial Practice You Need to Understand the Real Business World

IN THIS VOLUME

- Part X — Investment, Capital Markets and Asset Management
- Part XI — Commodities, Energy and Physical Trading
- Part XXI — How to Read Business News and Reports

A practical reference for finance and computer-science graduates entering the business world · First edition — October 2026

About this volume

This volume is part of a ten-volume reference collection. Chapter numbers run continuously across the collection, so a reference such as “§47.3” identifies the same section in every volume, in the master index (Volume 10), and in the learning roadmap.

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Vol. 1	Foundations and the Language of Business
Vol. 2	How Businesses Work — Accounting and Financial Statements
Vol. 3	Corporate Finance, Economics and Business Mathematics
Vol. 4	Banking, International Trade and Treasury
Vol. 5	Commercial Operations — Sales, Supply Chain, Insurance and Risk
Vol. 6	**Investment, Capital Markets, Commodities and Market Intelligence**
Vol. 7	Law, Entrepreneurship, Management and Technology
Vol. 8	Professional Practice — Communication, Deals and Advanced Thinking
Vol. 9	Document Library, Master Reference, Exercises and Learning Roadmap
Vol. 10	Master Table of Contents, Glossary and Unified Index

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PART X

Investment, Capital Markets and Asset Management

الاستثمار وأسواق رأس المال وإدارة الأصول

Capital markets connect those who have capital with those who need it, and price the risk in between. For a business professional, understanding investment is useful three times over: to raise capital for a company, to evaluate companies as a lender, partner or acquirer, and to manage personal and institutional wealth. This part explains the structure of equity and debt markets, the main investment vehicles and managers, portfolio construction, derivatives, and how professional investors research and decide.

LEARNING OBJECTIVES · Learning objectives for Part X

- Explain primary and secondary markets, IPOs, and the roles of exchanges, brokers and clearing houses.
- Analyse equities with market capitalisation, multiples and dividend measures.
- Price bonds, compute yields and duration, and interpret credit ratings and yield curves.
- Distinguish mutual funds, ETFs, private equity, venture capital and hedge funds.
- Apply risk-return, diversification and asset-allocation principles; compute risk-adjusted returns.
- Explain futures, forwards, options and swaps; distinguish investing, trading, speculation, hedging and arbitrage.
- Conduct investment research and due diligence and write an investment recommendation.

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CHAPTER 78

Capital Markets: Structure, Primary and Secondary Markets, and IPOs

LEARNING OBJECTIVES · In this chapter

How capital markets are organised, who the participants are, how securities are issued and traded, and how an initial public offering works.

78.1 Market structure

Market	Instruments	Main participants
Equity markets	Ordinary and preference shares, depositary receipts	Issuers, institutional and retail investors, brokers, market makers, exchanges
Debt (fixed income) markets	Government bonds, corporate bonds, sukuk, commercial paper, securitised products	Governments, corporates, banks, pension funds, insurers
Money markets	Treasury bills, certificates of deposit, repos, commercial paper (< 1 year)	Banks, corporates, money-market funds, central banks
Derivatives markets	Futures, options, swaps, forwards	Hedgers, speculators, arbitrageurs, dealers
FX markets	Spot, forwards, swaps, options	Banks, corporates, investors
Commodities markets	Futures, physical	Producers, consumers, traders

Primary market: issuers sell new securities to investors and receive the cash. **Secondary market:** investors trade existing securities among themselves; the issuer receives nothing, but secondary liquidity makes primary issuance possible and provides price signals.

78.2 Market infrastructure

- **Exchanges** (e.g., Saudi Exchange (Tadawul), Dubai Financial Market, Abu Dhabi Securities Exchange, Boursa Kuwait, Qatar Stock Exchange, the Egyptian Exchange, NYSE, Nasdaq, London Stock Exchange) provide organised trading with listing rules.
- **Over-the-counter (OTC) markets:** bilateral trading through dealers (most bonds, FX, many derivatives).
- **Brokers** execute orders for clients; **dealers/market makers** trade from their own inventory, quoting bid and ask.
- **Central counterparties (CCPs)/clearing houses** interpose themselves between buyers and sellers, reducing counterparty risk through margining.
- **Central securities depositories (CSDs)** hold securities electronically and settle transfers (delivery versus payment).
- **Custodians** safekeep assets for investors.
- **Regulators** (securities commissions, capital market authorities) supervise disclosure, conduct and market integrity.

78.3 Order types and trading

Market orders (execute now at the best price), limit orders (execute only at a set price or better), stop orders (trigger when a price is reached). Liquidity is measured by bid-ask spread, depth and volume. Settlement cycles: many equity markets settle on T+2; the US moved to **T+1** in May 2024 and other markets are planning similar moves.

78.4 Initial public offerings (IPOs)

Phase	Activities
Preparation (6–18 months)	Governance (independent directors, committees), audited IFRS statements (typically 3 years), reporting systems, legal restructuring, choice of exchange
Advisers	Lead managers/bookrunners, legal counsel, auditors, IR advisers, receiving banks
Due diligence and prospectus	Business, financial and legal due diligence; prospectus approved by regulator
Valuation and price range	Comparable companies, DCF; investor education by analysts; anchor/cornerstone investors
Bookbuilding	Institutional orders at different prices; retail tranche in many Middle Eastern IPOs
Pricing and allocation	Final price set; allocations to investors
Listing and trading	First trading day; stabilisation (greenshoe option) in some markets
Post-IPO	Continuous disclosure, quarterly reporting, investor relations, lock-up periods for existing shareholders

Why companies go public: raise growth capital, provide an exit for founders/investors, currency for acquisitions (shares), visibility and credibility, employee incentives. Costs: fees (often several percent of proceeds), disclosure burden, short-term market pressure, loss of privacy and some control.

بالعربية — الأسواق الأولية والثانوية والطرح العام الأولي

في السوق الأولية تبيع الشركة أوراقاً مالية جديدة وتحصل على الأموال، أما في السوق الثانوية فيتداول المستثمرون الأوراق القائمة فيما بينهم دون أن تحصل الشركة على شيء، لكن سيولة السوق الثانوية هي ما يجعل الإصدار الأولي ممكناً. والطرح العام الأولي (IPO) هو أول بيع لأسهم الشركة للجمهور، ويتطلب حوكمة قوية وقوائم مالية مدققة ونشرة إصدار معتمدة من الجهة الرقابية وعملية بناء سجل الأوامر لتحديد السعر.

PRACTICE · Practice questions

1. A company sells 20 million new shares at $10 in an IPO, while existing shareholders sell 10 million shares. How much does the company receive (before fees)?
2. Why do IPO investors care about lock-up periods?
3. Distinguish a broker from a market maker.

ANSWER · Answers to the practice questions

1. **$200m** (primary shares only). The $100m from secondary shares goes to selling shareholders.
2. When lock-ups expire, insiders can sell large volumes, potentially pressuring the share price; lock-ups also signal insiders' commitment.

3. A broker acts as agent, executing client orders for a commission; a market maker acts as principal, buying and selling from its own inventory and earning the bid-ask spread while providing liquidity.

CHAPTER 79

Equity Investing: Market Capitalisation, Valuation Multiples and Dividends

LEARNING OBJECTIVES · In this chapter

What shareholders own, sources of equity returns, valuation multiples and their interpretation, dividend investing, and how to read a stock in a structured way.

79.1 Sources of equity returns

FORMULA · Total shareholder return

TSR = (Price_end − Price_start + Dividends) ÷ Price_start
Long-run decomposition (approximate): Return ≈ Dividend yield + Earnings growth + Change in valuation multiple

Over long periods, earnings growth and dividends drive most returns; changes in multiples dominate short periods.

79.2 Market capitalisation and classification

Market cap = share price × shares outstanding. Categories (conventional, vary by market): large cap, mid cap, small cap, micro cap. **Free float** — shares available for public trading (excluding strategic and government holdings) — matters for index weights and liquidity; many Gulf companies have large government or family holdings and limited free float.

79.3 Valuation multiples

Multiple	Formula	Use	Caution
P/E	Price ÷ EPS	Most quoted; profitable companies	Distorted by one-offs, leverage, accounting
Forward P/E	Price ÷ next-year EPS estimate	Market expectations	Depends on forecasts
PEG	P/E ÷ EPS growth rate (%)	Adjust P/E for growth	Growth estimates uncertain
EV/EBITDA	EV ÷ EBITDA	Comparing across capital structures	Ignores capex and working capital
EV/EBIT	EV ÷ EBIT	Capital-intensive firms	—
EV/Sales	EV ÷ revenue	Loss-making or early-stage companies	Ignores profitability
P/B	Price ÷ book value per share	Banks, insurers, asset-heavy firms	Book values may be stale
Dividend yield	DPS ÷ price	Income investors	High yield may signal risk
FCF yield	FCF ÷ market cap (or EV)	Cash generation	FCF volatile

WORKED EXAMPLE · Reading two fictional listed companies

	Company P (logistics)	Company Q (software)
Share price	$24.00	$60.00
EPS	$2.00	$1.20
P/E	12.0×	50.0×
Expected EPS growth	6%	30%
PEG	2.0	1.7
Dividend yield	4.5%	0%
Net debt/EBITDA	2.5×	net cash

Q looks “expensive” on P/E but cheaper on PEG; P offers income but slower growth and more debt. The right choice depends on confidence in Q’s growth, the investor’s income needs and risk tolerance.

79.4 Dividends

Dividend investing focuses on stable, growing dividends. Key measures: **payout ratio** (dividends ÷ earnings), **dividend cover** (earnings ÷ dividends), **FCF cover**, dividend history and policy. Warning signs: payout above 100% of earnings or FCF, dividends funded by debt, a yield far above peers (the market may expect a cut).

FORMULA · Dividend discount model (Gordon growth)

$P_0 = D_1 \div (Ke - g)$
Implied cost of equity: $Ke = D_1 \div P_0 + g$

Example: next dividend $1.20, expected growth 4%, required return 10% → value = 1.20 ÷ 0.06 = **$20.00**.

79.5 A structured approach to analysing a stock

1. **Business quality:** model, moat, industry structure, management, capital allocation record.
2. **Financials:** growth, margins, ROIC vs WACC, cash conversion, balance sheet.
3. **Valuation:** multiples vs history and peers, DCF, implied expectations (“what growth does the price assume?”).
4. **Catalysts and risks:** what could change the market’s view; what could go wrong.
5. **Position sizing and exit criteria.**

PRACTICE · Practice questions

1. Bought a share at $50, received $2 of dividends, sold at $56 after one year. Compute TSR and its components.
2. A stock trades at $40 with expected dividend next year $1.60 and long-term growth 5%. What cost of equity is implied?
3. Why is a P/E ratio not meaningful for a bank-like comparison with an industrial company?

ANSWER · Answers to the practice questions

1. TSR = (56 − 50 + 2) ÷ 50 = **16%**: price return 12%, dividend yield 4%.
2. Ke = 1.60 ÷ 40 + 5% = **9%**.
3. P/E itself can be computed for both, but comparing it across very different business models, leverage levels and accounting (banks’ earnings depend on provisions and leverage; industrials on

> capex cycles) is misleading; compare within sectors and use sector-specific metrics (P/B and ROE for banks; EV/EBITDA for industrials).

CHAPTER 80

Fixed Income: Bond Pricing, Yields, Credit Ratings and the Yield Curve

LEARNING OBJECTIVES · In this chapter

How bonds work, how they are priced, yield measures, duration and convexity, credit ratings and spreads, the yield curve, and sukuk.

80.1 Bond terms

Term	Meaning
Face value / par	Amount repaid at maturity (e.g., $1,000 or 100%)
Coupon	Annual interest as % of par; paid annually or semi-annually
Maturity	Date of final repayment
Issue price	Price at issuance (at, above or below par)
Yield to maturity (YTM)	Discount rate equating the present value of cash flows to the price
Seniority / security	Ranking in default (senior secured, senior unsecured, subordinated)
Covenants	Restrictions on the issuer (often incurrence-based for high-yield bonds)
Call / put options	Issuer may redeem early / investor may demand early repayment
Accrued interest	Interest since last coupon; buyer pays seller "dirty price" = clean price + accrued

80.2 Pricing

FORMULA · Bond price

$$\text{Price} = \Sigma\,[C \div (1+y)^t] + F \div (1+y)^n$$

C = coupon payment; F = face value; y = yield per period; n = number of periods.
Price and yield move in opposite directions.

WORKED EXAMPLE · A 5-year 6% annual-coupon bond

- If market yield = 6% → price = **100.00** (par).
- If yield = 7% → price = 6 × 4.1002 + 100 ÷ 1.40255 = 24.60 + 71.30 = **95.90** (discount).
- If yield = 5% → price = 6 × 4.3295 + 100 ÷ 1.27628 = 25.98 + 78.35 = **104.33** (premium).
- Current yield at 95.90 = 6 ÷ 95.90 = **6.26%**.

80.3 Duration and convexity

Macaulay duration is the weighted-average time to receive the bond's cash flows (weights = PV of each cash flow ÷ price). **Modified duration** ≈ Macaulay ÷ (1 + y) measures price sensitivity: %ΔPrice ≈ −Modified duration × Δy.

WORKED EXAMPLE · Duration of the 6% bond at 7%

PVs of cash flows: 5.61, 5.24, 4.90, 4.58, 75.58 (total 95.90). Weighted times: Σ t × PV = 426.97 → Macaulay duration = 426.97 ÷ 95.90 = **4.45 years**; modified duration = 4.45 ÷ 1.07 = **4.16**. If yields rise 1 point to 8%, estimated price change ≈ −4.16% → 91.91; actual price at 8% = **92.01** — the small difference is **convexity** (prices fall less and rise more than duration predicts).

Practical uses: banks and insurers match asset and liability durations; bond investors adjust duration according to rate views; companies understand why long-term fixed-rate debt changes in market value.

80.4 Credit ratings and spreads

Rating agencies (S&P Global Ratings, Moody's, Fitch, and regional agencies) assess default risk.

Category	S&P/Fitch	Moody's	Meaning
Investment grade	AAA, AA, A, BBB	Aaa, Aa, A, Baa	Lower default risk; eligible for many institutional mandates
High yield (speculative)	BB, B, CCC and below	Ba, B, Caa and below	Higher default risk; higher spreads
Default	D (S&P), D/RD (Fitch)	C (often)	In default

A bond's yield = benchmark government yield (risk-free in that currency) + **credit spread** (compensation for default risk, liquidity and other factors). Spreads widen in recessions and crises. A downgrade from BBB− to BB+ ("fallen angel") can force sales by investment-grade-only investors.

80.5 The yield curve

The yield curve plots yields by maturity for the same issuer (usually the government). Shapes: **normal** (upward), **flat**, **inverted** (downward — often associated with expected rate cuts and slowing growth), **humped**. Theories: expectations of future short rates, term premia for longer maturities, and supply/demand segments.

80.6 Sukuk

Sukuk are Sharia-compliant certificates giving holders ownership interests in assets or ventures (ijara sukuk based on leases, murabaha, wakala, mudaraba structures). Economically many resemble bonds (periodic distributions, redemption at maturity), and they are significant in Gulf and Malaysian markets for governments, banks and corporates. Structural features (asset ownership, purchase undertakings) affect credit analysis.

PRACTICE · Practice questions

1. Price a 3-year 8% annual-coupon bond (face 100) at a yield of 6%.
2. A bond has modified duration 6.5. Yields rise by 0.40 percentage points. Estimate the price change.
3. A BBB corporate bond yields 6.1% when the same-maturity government bond yields 4.3%. What is the spread in basis points? What might cause it to widen?

ANSWER · Answers to the practice questions

1. 8 ÷ 1.06 + 8 ÷ 1.1236 + 108 ÷ 1.191016 = 7.547 + 7.120 + 90.679 = **105.35**.
2. ≈ −6.5 × 0.40% = **−2.6%**.

3. **180 bps**. Widening causes: weaker issuer results, downgrade risk, sector stress, recession fears, lower market liquidity, higher risk aversion.

CHAPTER 81

Investment Funds and Asset Managers: Mutual Funds, ETFs, Private Equity, Venture Capital and Hedge Funds

LEARNING OBJECTIVES · In this chapter

How pooled investment vehicles work, their fee structures, liquidity, strategies and roles; what investment mandates are.

81.1 Comparison of vehicles

Vehicle	Investors	Liquidity	Typical strategy	Fees (indicative)	Regulation
Mutual fund (open-ended)	Retail and institutional	Daily at NAV	Equities, bonds, money market, balanced	Management fee 0.5–2% (index funds lower)	Heavily regulated (e.g., UCITS in the EU)
ETF	Retail and institutional	Intraday on exchange	Mostly index tracking; some active	Often 0.03–0.75%	Regulated; market makers keep price near NAV
Closed-end fund / REIT	Retail and institutional	Exchange-traded; price may differ from NAV	Real estate, credit, private assets	Varies	Regulated
Private equity fund	Institutions, wealthy individuals	Locked ~10 years	Buyouts, growth equity	"2 and 20": ~1.5–2% management fee + ~20% carried interest above a hurdle	Lighter; professional investors
Venture capital fund	Institutions, family offices	Locked ~10+ years	Early-stage startups	Similar to PE	Professional investors
Hedge fund	Institutions, wealthy individuals	Monthly/quarterly with notice; lock-ups	Long/short, macro, arbitrage, credit, quant	Management 1–2% + performance 10–20%	Lighter; professional investors
Private credit fund	Institutions	Locked	Direct lending to mid-sized companies	Management + performance	Professional investors
Sovereign wealth fund	Government-owned	Long-term	Global diversified, strategic	—	Government mandates

81.2 Key concepts

- **NAV (net asset value) per unit** = (assets − liabilities) ÷ units outstanding.
- **Active vs passive:** active managers try to beat a benchmark; passive funds track it at low cost. After fees, many active funds underperform their benchmarks over long periods — a key argument for passive investing in efficient markets.
- **Private equity mechanics:** limited partners (LPs) commit capital; the general partner (GP) calls capital over an investment period, invests, manages portfolio companies, exits (sale, IPO), and distributes proceeds. Performance measures: IRR, MOIC/TVPI (total value to paid-in), DPI (distributions to paid-in). The **J-curve**: early negative returns from fees and unrealised investments, followed by gains.
- **Venture capital:** power-law returns — a few investments generate most of the fund's returns.

- **Hedge fund strategies:** equity long/short, global macro, event-driven (merger arbitrage), relative-value/arbitrage, credit, quantitative/systematic, multi-strategy.

81.3 Investment mandates

A mandate specifies objectives (return targets, income), constraints (risk limits, asset classes, ratings, concentration, liquidity, leverage, currency, ESG or Sharia screening), benchmark, reporting and fees. Example: "Global investment-grade bond portfolio; benchmark Bloomberg Global Aggregate (hedged to USD); max 10% BBB; no derivatives except currency forwards for hedging; tracking error ≤ 1.5%."

WORKED EXAMPLE · Private equity fees and carried interest (simplified, fictional)

Fund of $100m; management fee 2% per year for 5 years then 1.5% on invested capital; hurdle 8% preferred return; carry 20% with catch-up. Over 8 years, total fees ≈ $14m; investments ($86m) return $190m. Profit to distribute = 190 − 100 = $90m (after returning all contributed capital). With full catch-up, the GP receives 20% of the total profit = **$18m**; LPs receive **$72m** profit — net MOIC ≈ 1.72× and net IRR roughly in the low teens, depending on timing.

PRACTICE · Practice questions

1. A fund has assets of $520m, liabilities $20m and 40m units. Compute NAV per unit.
2. Why can an ETF's market price differ slightly from its NAV, and what keeps the gap small?
3. Explain the J-curve to a new investor in a PE fund.

ANSWER · Answers to the practice questions

1. (520 − 20) ÷ 40 = **$12.50**.
2. ETF shares trade continuously on the exchange; supply and demand move the price. Authorised participants can create or redeem ETF units in exchange for the underlying basket, arbitraging away large differences.
3. In the early years the fund charges fees and holds investments at cost or below while new companies are being improved, so reported returns are negative; as investments mature and are sold, distributions and valuations rise, and returns turn positive — tracing a "J" shape.

CHAPTER 82

Portfolio Management: Risk, Return, Diversification and Asset Allocation

LEARNING OBJECTIVES · In this chapter

Expected return and risk, diversification and correlation, asset allocation, risk-adjusted performance measures, and practical risk measures such as drawdown and value at risk.

82.1 Risk and return

Investors demand higher expected returns for bearing more risk. Risk can be measured as volatility (standard deviation of returns), downside deviation, maximum drawdown, probability of loss, or — for long-term investors — the risk of failing to meet objectives.

FORMULA · Two-asset portfolio

$E(R_p) = w_1E(R_1) + w_2E(R_2)$

$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{12}\sigma_1\sigma_2$

Diversification benefit is larger when correlation ρ is low.

WORKED EXAMPLE · A 60/40 portfolio (illustrative assumptions)

Equities: expected return 9%, volatility 18%. Bonds: 4.5%, volatility 6%. Correlation 0.2. Risk-free rate 4%.

- $E(R_p) = 0.6 \times 9 + 0.4 \times 4.5 =$ **7.2%**.
- $\sigma_p^2 = 0.36 \times 0.0324 + 0.16 \times 0.0036 + 2 \times 0.6 \times 0.4 \times 0.2 \times 0.18 \times 0.06 = 0.013277 \rightarrow \sigma_p =$ **11.5%** (less than the weighted average volatility of 13.2%).
- Sharpe ratio = $(7.2 - 4.0) \div 11.5 =$ **0.28**.

82.2 Systematic and idiosyncratic risk

Idiosyncratic (company-specific) risk can be diversified away by holding many securities; **systematic** (market) risk cannot. CAPM states that only systematic risk (beta) is rewarded with higher expected returns. Practitioners also recognise other systematic factors (size, value, momentum, quality, low volatility) used in factor investing.

82.3 Asset allocation

Strategic asset allocation — the long-term mix of asset classes — explains most of the variability of a diversified portfolio's returns over time. It depends on the investor's:

Factor	Example effect
Objectives	Growth vs income vs capital preservation
Time horizon	Longer horizon → more equities, illiquid assets
Risk tolerance (ability and willingness)	Lower tolerance → more bonds and cash
Liquidity needs	Pension payouts, business capital needs

Factor	Example effect
Constraints	Regulation, Sharia compliance, ESG policy, currency of liabilities
Tax	Location and choice of vehicles

Tactical asset allocation makes short-term deviations; **rebalancing** restores target weights, systematically selling assets that have risen and buying those that have fallen.

> **WORKED EXAMPLE · Investment policy for a family business's surplus cash (fictional)**
>
> Surplus $20m not needed for 5+ years, plus $5m liquidity reserve. Policy: liquidity reserve in deposits and T-bills (≤ 12 months, A-rated banks, ≤ 25% per bank). Long-term pool: 50% global equities (index funds), 30% investment-grade bonds/sukuk (duration 3–6 years), 10% real estate (listed REITs), 10% private markets via diversified funds; rebalance when weights deviate by more than 5 points; no single security above 5%; no leverage.

82.4 Risk-adjusted performance measures

Measure	Formula	Interpretation
Sharpe ratio	(Rp − Rf) ÷ σp	Excess return per unit of total risk
Sortino ratio	(Rp − Rf) ÷ downside deviation	Penalises only downside volatility
Treynor ratio	(Rp − Rf) ÷ βp	Excess return per unit of systematic risk
Jensen's alpha	Rp − [Rf + βp(Rm − Rf)]	Return above CAPM expectation
Information ratio	(Rp − Rb) ÷ tracking error	Active return per unit of active risk
Maximum drawdown	Largest peak-to-trough fall	Pain experienced by investors

82.5 Value at risk (VaR)

VaR estimates the maximum loss over a horizon at a confidence level under normal conditions. Parametric example: portfolio $10m, annual volatility 12% → daily σ ≈ 12% ÷ √252 = 0.756%; 95% one-day VaR ≈ 1.65 × 0.756% × $10m ≈ **$125,000**. VaR says nothing about losses beyond the threshold; **stress tests** and **expected shortfall** complement it.

> **PRACTICE · Practice questions**
>
> 1. Two assets: A (return 10%, σ 20%), B (return 5%, σ 8%), correlation −0.1. Compute expected return and volatility of a 50/50 portfolio.
> 2. Portfolio return 11%, benchmark 9%, tracking error 4%. Information ratio?
> 3. Portfolio return 12%, beta 1.2, risk-free 4%, market return 10%. Compute Jensen's alpha.

> **ANSWER · Answers to the practice questions**
>
> 1. E = **7.5%**; σ^2 = 0.25 × 0.04 + 0.25 × 0.0064 + 2 × 0.25 × (−0.1) × 0.2 × 0.08 = 0.01 + 0.0016 − 0.0008 = 0.0108 → σ = **10.4%**.
> 2. (11 − 9) ÷ 4 = **0.5**.
> 3. Expected = 4 + 1.2 × (10 − 4) = 11.2% → alpha = **+0.8%**.

CHAPTER 83

Derivatives: Forwards, Futures, Options and Swaps

LEARNING OBJECTIVES · In this chapter

How the main derivatives work, their payoffs, margining and valuation intuition; and the differences between investing, trading, speculation, hedging and arbitrage, with numerical examples.

83.1 The four building blocks

Instrument	Definition	Traded	Obligation	Upfront cost	Main risk
Forward	Agreement to buy/sell at a future date at a fixed price	OTC	Both parties	None (credit lines/collateral)	Counterparty
Future	Standardised forward traded on an exchange, cleared, marked to market daily	Exchange	Both	Initial margin	Margin calls (liquidity)
Option	Right (not obligation) to buy (call) or sell (put) at a strike price	Exchange or OTC	Seller (writer) only	Premium paid by buyer	Buyer: premium; seller: potentially large losses
Swap	Exchange of cash-flow streams (fixed vs floating rates, currencies, commodity prices)	OTC (many cleared)	Both	Usually none	Counterparty, market value changes

83.2 Futures margining

WORKED EXAMPLE · Crude oil futures (illustrative)

A refiner buys 10 futures contracts (1,000 barrels each) at $70.00 to hedge future purchases. Initial margin $6,000 per contract = $60,000.

- Day 1 settlement $68.50 → loss 1.50 × 10,000 = **$15,000** variation margin paid.
- Day 2 settlement $71.20 → gain 2.70 × 10,000 = **$27,000** received. Daily mark-to-market eliminates most counterparty risk but creates **liquidity risk**: a hedger may need cash for margin calls before the physical benefit arrives.

83.3 Option payoffs

FORMULA · Option payoffs at expiry

Long call: $\max(S - K, 0) - \text{premium}$
Long put: $\max(K - S, 0) - \text{premium}$
Short call: $\text{premium} - \max(S - K, 0)$
Short put: $\text{premium} - \max(K - S, 0)$
Break-even (long call) = $K + \text{premium}$; (long put) = $K - \text{premium}$

WORKED EXAMPLE · Call option on a share

Strike $100, premium $5. At expiry: S = 90 → payoff −5 (premium lost); S = 105 → 0 (break-even); S = 120 → +15. The buyer's loss is limited to $5; the seller's potential loss is unlimited for an uncovered call.

Option value depends on the underlying price, strike, time to expiry, volatility, interest rates and dividends (Black–Scholes and binomial models). Volatility is the key input — options are, in effect, priced volatility.

83.4 Swaps

- **Interest-rate swap:** pay fixed, receive floating (or vice versa) on a notional amount. Used by borrowers to fix rates (see Part XVIII) and by banks to manage interest-rate risk.
- **Cross-currency swap:** exchange principal and interest in two currencies.
- **Commodity swap:** pay fixed price, receive floating index price (or vice versa) for a quantity — used by airlines (jet fuel), producers, consumers.
- **Credit default swap (CDS):** protection buyer pays a premium; seller pays if the reference entity defaults.

ISDA documentation (master agreement, schedule, credit support annex) governs most OTC derivatives.

83.5 Investing, trading, speculation, hedging and arbitrage

Activity	Purpose	Time horizon	Source of return / benefit	Example
Investing	Own productive assets for long-term returns	Years	Cash flows (dividends, interest) and growth	Buy a diversified equity index fund
Trading	Profit from short-term price movements and liquidity	Seconds to months	Price changes, spreads	Market-making in bonds
Speculation	Take risk on a view of future prices	Varies	Correct forecasts	Buying oil futures expecting a price rise without any oil exposure
Hedging	Reduce an existing risk	Matches exposure	Stability of cash flows and margins	Airline buys jet-fuel swaps
Arbitrage	Exploit price differences for the same or equivalent asset with little or no risk	Very short	Convergence of prices	Buying a commodity spot and selling futures when the spread exceeds carry costs

WORKED EXAMPLE · Same instrument, three purposes

Wheat futures at $6.00/bushel.

- **Hedger:** a flour mill that must buy wheat in 3 months buys futures; if wheat rises to $7.00, the futures gain offsets the higher physical cost.
- **Speculator:** a fund with no wheat exposure buys futures expecting a drought; it gains if prices rise, loses if they fall.
- **Arbitrageur:** if spot wheat is $5.40 and 3-month futures $6.00 while storage, insurance and financing cost $0.35, it buys spot, stores, and sells futures, locking in about $0.25 per bushel (less transaction costs and basis risks).

83.6 Market liquidity and volatility

Liquidity determines how easily positions are opened or closed without moving prices; it can vanish in stress. Volatility drives option prices, margin requirements and VaR. Volatility indices (e.g., VIX for the S&P 500) measure implied volatility expectations.

COMMON MISTAKES · Derivatives risks for companies

- Leverage: small price moves create large gains/losses relative to cash invested.
- Liquidity: margin calls and collateral requirements.
- Basis risk: hedges do not move exactly with the exposure.
- Counterparty risk in OTC contracts.
- Complexity: structured products can contain sold options with hidden downside.
- Accounting volatility without hedge accounting.
- Governance: unauthorised trading, poor controls.

PRACTICE · Practice questions

1. A put option with strike \$50 costs \$3. Compute payoff at expiry prices \$40, \$47, \$55 and the break-even.
2. An airline will buy 1 million gallons of jet fuel in 6 months. Which derivative positions could hedge this, and what is the trade-off between a swap and a call option?
3. Classify: (a) a pension fund buying government bonds to match pension liabilities; (b) a trader buying gold in London and simultaneously selling it at a higher price in Dubai; (c) a CFO buying USD forward although the company has no USD payments, because she expects the local currency to weaken.

ANSWER · Answers to the practice questions

1. S = 40: 10 − 3 = **+7**; S = 47: 3 − 3 = **0**; S = 55: 0 − 3 = **−3**. Break-even **\$47**.
2. Pay-fixed jet-fuel swap (or long futures in a related contract such as heating oil/gasoil, with basis risk) or buy call options/caps. Swap: no premium, locks price, but no benefit if prices fall. Call: premium cost, protects against rises while keeping the benefit of falls.
3. (a) Investing/hedging liabilities (asset-liability matching); (b) arbitrage (if simultaneous, with costs and logistics accounted for); (c) speculation — not permitted under a typical treasury policy.

CHAPTER 84

Investment Research, Due Diligence and a Fictional Investment Case

LEARNING OBJECTIVES · In this chapter

How professional investors research opportunities, conduct due diligence, and write an investment memorandum — illustrated with a complete fictional case.

84.1 The research process

1. **Idea generation:** screens (valuation, quality, momentum), industry themes, events (spin-offs, IPOs), contrarian situations.
2. **Business understanding:** annual reports, industry studies, customer and supplier conversations, competitor analysis, management meetings.
3. **Financial analysis:** historical normalisation, drivers, quality of earnings, balance sheet strength.
4. **Valuation:** multiple methods; reverse-engineering market expectations.
5. **Thesis and variant perception:** why the market is wrong, and what evidence will prove it.
6. **Risk assessment:** downside scenarios, what would make the thesis wrong (pre-mortem).
7. **Decision and sizing:** expected return vs risk; position size limited by conviction and liquidity.
8. **Monitoring:** milestones, thesis review, exit discipline.

84.2 Due diligence for investments

Area	Questions
Business and market	Is demand durable? What is the competitive position and moat?
Management and governance	Track record, incentives, related-party transactions, board independence
Financial	Accuracy of reported figures, cash conversion, off-balance-sheet items, accounting policies
Legal and regulatory	Licences, litigation, compliance history
ESG	Environmental liabilities, labour practices, governance quality
For funds (manager due diligence)	Strategy consistency, team stability, track record attribution, risk management, operations, fees and terms, alignment of interests

84.3 Fictional investment case: Gulf Cold Chain Holdings (listed)

CASE STUDY · Investment memorandum (training simulation)

Company: Gulf Cold Chain Holdings (fictional), listed; operates temperature-controlled warehouses and refrigerated trucking in three Gulf countries. **Share price:** $8.00 · **Shares:** 250m · **Market cap:** $2.0bn · **Net debt:** $500m (including leases) · **EV:** $2.5bn.

$m	FY-2	FY-1	FY0	FY+1E	FY+2E
Revenue	380	425	470	530	595
EBITDA	110	125	140	162	185
EBITDA margin	28.9%	29.4%	29.8%	30.6%	31.1%

$m	FY-2	FY-1	FY0	FY+1E	FY+2E
Capex	120	150	160	140	120
Free cash flow	(40)	(55)	(45)	0	35
EPS ($)	0.20	0.23	0.26	0.30	0.36

Valuation: EV/EBITDA FY+1E = 2,500 ÷ 162 = **15.4×**; peers (global cold-chain operators) 12–16×; P/E FY+1E = 26.7×.

Thesis: (1) structural growth in chilled and frozen food and pharma distribution, driven by population growth, food-security policies and e-grocery; (2) high switching costs and limited licensed cold-storage capacity in prime locations; (3) capex peak passing — FCF turning positive in FY+2, enabling dividends and deleveraging.

What the price implies: reverse DCF at WACC 9% suggests the market assumes ~10% revenue growth for 10 years and stable margins — in line with management guidance, so the upside depends on margin expansion and capital discipline.

Risks: new capacity from competitors and government-backed logistics zones; customer concentration (top 5 = 35% of revenue); energy costs; execution of capex projects; leverage at 3.6× net debt/EBITDA.

Scenarios (12-month target):

Scenario	Probability	FY+1 EBITDA	Multiple	Implied price
Bull	25%	170	16×	$8.88
Base	50%	162	14.5×	$7.40
Bear	25%	145	11×	$4.38
Probability-weighted				**$7.02**

(Implied price = (EBITDA × multiple − net debt $500m) ÷ 250m shares.)

Recommendation: Do not buy at $8.00 — the expected value ($7.02) is below the price, and the bear case shows meaningful downside given leverage. Revisit if the price falls below ~$6.50 or if evidence emerges that margins can exceed 32% (e.g., pharma contracts).

Training simulation — fictional company and figures.

PROFESSIONAL INSIGHT · Graduate vs professional

The graduate likes the growth story and concludes “buy”. The professional asks what growth is already in the price, how leverage amplifies the bear case, whether free cash flow will actually arrive, and what the expected value is relative to the price.

PRACTICE · Practice questions

1. Recompute the implied price in the base case if net debt were $300m instead of $500m.
2. What three pieces of evidence would make you change the recommendation to “buy”?
3. List five due-diligence questions you would ask management.

ANSWER · Answers to the practice questions

1. (162 × 14.5 − 300) ÷ 250 = (2,349 − 300) ÷ 250 = **$8.20**.

2. Signed long-term pharma contracts at higher margins; capex guidance cut with no growth loss (FCF earlier); occupancy data showing pricing power (rate increases accepted); share price fall that leaves the base case with upside.
3. What is the contract length and renewal rate of top customers? What are utilisation rates by site and pricing trends? What return do you expect on new warehouses and how have past projects performed? How are energy costs managed/hedged? What is the capital allocation priority between growth, dividends and debt reduction?

SUMMARY · Part X summary

- Primary markets raise capital; secondary markets provide liquidity and pricing.
- Equity returns come from dividends, earnings growth and multiple changes; compare multiples within sectors.
- Bond prices move inversely to yields; duration measures sensitivity; spreads price credit risk.
- Funds differ in liquidity, strategy, fees and investor eligibility; mandates set the rules.
- Diversification reduces idiosyncratic risk; asset allocation drives portfolio outcomes; measure performance on a risk-adjusted basis.
- Derivatives can hedge, speculate or arbitrage — purpose and governance matter.
- Good investment decisions compare expected value with price and respect the downside.

PART XI

Commodities, Energy and Physical Trading

السلع والطاقة والتجارة الفعلية

Physical commodity trading moves the raw materials of the world economy — oil, gas, grain, sugar, coffee, metals, ore, stone and timber — from producers to consumers across oceans and borders. It looks simple ("buy low, sell high") but is one of the most operationally and financially demanding businesses: margins are thin, volumes are large, prices are volatile, logistics are complex, and a single failed cargo can wipe out a year's profit.

This part explains how physical trading works in general, then covers energy, agricultural, metal and industrial commodities individually, and ends with three complete fictional deal P&Ls. **All prices, differentials, freight rates and margins in this part are illustrative assumptions for teaching, not market data.**

LEARNING OBJECTIVES · **Learning objectives for Part XI**

- Explain the roles of producers, consumers, brokers, traders and trading houses.
- Understand benchmarks, pricing formulas (flat price, benchmark ± differential, pricing periods), and quality and quantity determination.
- Explain spot, forward and futures transactions, basis risk, physical premiums, spreads and freight economics.
- Describe contracts, inspection, storage, financing, insurance and delivery obligations for each major commodity group.
- Compute the full profitability of a physical commodity deal.
- Distinguish physical trading from trading financial instruments.

Chapters in this part

Chapter 85 How Physical Commodity Trading Works 27
Chapter 86 Energy: Crude Oil, Refined Products, Natural Gas, LNG and Coal 32
Chapter 87 Agricultural Commodities: Wheat, Corn, Sugar and Coffee 34
Chapter 88 Metals: Aluminium, Copper, Gold and Iron Ore 36
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Chapter 90 Commodity Deal Economics: Three Complete Trade P&Ls 40

CHAPTER 85

How Physical Commodity Trading Works

LEARNING OBJECTIVES · In this chapter

The participants, sources of trading margin, pricing mechanisms, quality and quantity, logistics, contracts, hedging, financing, insurance and risks common to all physical commodities.

85.1 Participants

Participant	Role	Typical concerns
Producers	Mines, oil companies, farms and cooperatives, quarries, sawmills	Price realisation, offtake security, financing
End users / consumers	Refiners, smelters, mills, food processors, utilities, fabricators, contractors	Supply security, quality, cost, credit terms
Traders and trading houses	Buy, transport, store, blend and sell; manage risk and logistics	Margin, risk, financing, counterparty reliability
Brokers (physical)	Introduce buyers and sellers for commission; do not take title	Commission; avoiding circumvention
Brokers (futures/FCMs)	Execute exchange trades, hold margin	Client margin, compliance
Ship brokers	Fix vessels for charterers	Commission on freight
Inspection companies	Quantity/quality determination at load/discharge	Independence, accuracy
Collateral managers / warehouse operators	Control and certify stored goods for financiers	Integrity, insurance
Banks and trade financiers	Finance transactions, issue L/Cs, borrowing bases	Collateral control, fraud, KYC
Exchanges and price reporting agencies	Futures markets; published price assessments	Liquidity, methodology
Governments and state buyers	Tenders, export controls, strategic reserves	Food and energy security

85.2 Where does a trader's margin come from?

Source	Example
Location (geographical) arbitrage	Buy wheat FOB Black Sea, sell CFR North Africa when the price difference exceeds freight and costs
Time arbitrage	Buy and store when the forward curve is in contango
Quality arbitrage / blending	Blend crude or ore grades to meet buyer specifications at lower cost
Logistics optimisation	Better freight, vessel scheduling, combining cargoes
Credit and financing services	Selling on deferred terms to buyers who lack bank lines (priced into the premium)
Information and relationships	Knowing supply and demand flows earlier and better
Risk management	Managing price, basis and spread exposures intelligently

85.3 Pricing mechanisms

- **Flat (fixed) price:** a single agreed price per unit — creates full price exposure until hedged or sold.

- **Benchmark ± differential (premium/discount):** e.g., "average of Dated Brent quotations for the 5 days around B/L date + $1.10/bbl", or "LME cash settlement average for the month of shipment + $95/t". The differential reflects quality, location, timing and market tightness.
- **Pricing period (quotational period, QP):** the dates over which the benchmark is averaged, which may differ between purchase and sale — the gap is a price risk to hedge.
- **Formula pricing:** long-term contracts linking price to indices (e.g., LNG priced as a percentage of oil prices plus a constant).
- **Tenders:** state buyers publish requirements; sellers submit offers; lowest compliant offer usually wins.

85.4 Spot, forward, futures, basis and spreads

Concept	Meaning in physical trading
Spot / prompt	Delivery in the near term (days/weeks)
Forward (physical)	Contract for delivery at a future period at an agreed price or formula
Futures	Exchange contract used mostly for hedging and price discovery; few are physically delivered
Basis	Local physical price – futures price; reflects location, quality, timing and local supply/demand
Physical premium	Amount above the benchmark (e.g., regional aluminium premium)
Calendar spread	Price difference between delivery months (contango/backwardation)
Location spread	Price difference between regions (e.g., Brent vs Dubai)
Quality spread	Price difference between grades (e.g., high vs low protein wheat; sweet vs sour crude)
Crack spread / margin	Product prices minus crude cost (refining margin)
Freight economics	Location spreads must exceed freight + insurance + losses + financing for an arbitrage

WORKED EXAMPLE · Hedging a cargo and the basis

A trader buys 50,000 t of corn at a fixed $210/t FOB (equivalent to futures + $25 basis) and has not yet sold it. To remove flat-price risk, it sells futures equivalent to 50,000 t at $185. Two weeks later it sells the cargo at $201/t (futures have fallen to $180; basis has weakened to +$21).

- Physical: 201 − 210 = **−$9/t**.
- Futures hedge: 185 − 180 = **+$5/t**.
- Net: **−$4/t** = the change in basis (25 → 21). The hedge removed flat-price risk but not basis risk.

85.5 Quality and quantity

Contracts define **specifications** (e.g., protein, moisture and foreign matter for wheat; API gravity and sulphur for crude; Fe content and impurities for iron ore; purity for metals) and **adjustments** (price escalators and de-escalators for deviations, rejection limits). **Quantity** is determined by draft survey (ships), shore-tank or flow-meter measurements (oil), weighbridge (trucks), or count/area (slabs, logs), with an agreed **tolerance** (e.g., ±10% at seller's or buyer's option). Which certificate is **final** (loading or discharge, which inspector) is critical: in many commodity contracts quality and quantity are "final at loading", putting the risk of losses in transit on the buyer (who insures it).

85.6 Logistics, storage and inspection

Commodities move by bulk carriers, tankers, LNG carriers, containers, barges, rail and pipelines. Storage happens in tanks, silos, warehouses, stockyards and exchange-approved warehouses (for futures deliv-

ery). Independent inspectors certify quality and quantity; collateral managers control goods financed by banks.

85.7 Contracts

Standard industry contract forms reduce negotiation and disputes: e.g., **GAFTA** forms for grains and feed, **FOSFA** for oils and fats, **ICA** rules for cotton, **LME** rules for exchange metal, oil companies' **general terms and conditions** for crude and products, **BIMCO** charter-party forms (e.g., GENCON voyage charter) and tanker charter forms, and **ISDA** for OTC derivatives. Many forms include arbitration by the relevant trade association.

85.8 Hedging, financing and insurance

- **Hedging:** futures, swaps and options remove flat-price exposure; residual risks are basis, spread, quality and timing.
- **Financing:** transactional trade finance (L/Cs, import/export loans), borrowing-base facilities (on inventory and receivables), pre-export finance and prepayment to producers, repo of commodity stocks, receivables discounting. Banks typically finance 80–90% of the value of hedged, insured, controlled goods.
- **Insurance:** marine cargo (often required by banks), stock throughput insurance (covers goods in transit and storage), charterer's liability, political risk, credit insurance.

85.9 Delivery obligations

The seller must deliver the contracted quantity and quality within the delivery period, at the agreed place, with the required documents. Failure can trigger default clauses: the non-defaulting party may buy in or sell out against the defaulter and claim the price difference (market-based damages) under trade-association rules — a powerful mechanism that makes commodity contracts quite different from ordinary supply agreements.

85.10 The main risks

Risk	Description	Mitigation
Price (flat)	Price moves before purchase and sale are matched	Hedging, back-to-back pricing, limits
Basis and spread	Relationships between prices move	Monitoring, limits, diversified hedges
Counterparty / performance	Seller fails to deliver or buyer fails to pay or take delivery	KYC, credit limits, L/Cs, credit insurance, performance guarantees
Operational	Vessel delays, demurrage, contamination, quality failure, loss	Experienced operators, inspection, insurance
Fraud	Fake documents, non-existent goods, double financing	Due diligence, collateral managers, verification of B/Ls and warehouse receipts
Country and political	Export bans, sanctions, war, expropriation	Diversification, political risk insurance, compliance
Liquidity	Margin calls on hedges while physical cash comes later	Committed facilities, margin financing lines
Regulatory	Sanctions, market-abuse rules, ESG requirements	Compliance function

85.11 Typical margins and their limits

Gross margins in bulk trading are often a few dollars per tonne on grains or ores, cents per barrel to a few dollars on oil, and tens of dollars per tonne on metal premiums — a small percentage of the cargo value. Speciality products (e.g., natural stone, specialty coffees, niche chemicals) can offer higher percentage margins with lower volumes and higher operational risk. Published figures from public trading houses show that profits vary greatly with volatility and dislocations: margins tend to widen in disrupted markets and compress in calm ones. Any margin estimate must be stress-tested against freight, demurrage, quality claims, financing and counterparty risk.

85.12 Physical trading vs financial trading

	Physical trading	Financial trading (futures, swaps)
Object	Actual goods with quality, location, logistics	Standardised contracts
Profit sources	Location, time, quality arbitrage; logistics; credit services	Price movements, spreads, market-making
Operational complexity	High (vessels, inspection, storage, documents, customs)	Low
Capital	Working capital for goods, often bank-financed	Margin
Risks	Price, basis, performance, operational, fraud, country	Price, liquidity, counterparty (OTC)
Regulation	Trade, customs, sanctions, product rules; increasingly market regulation	Financial market regulation
Skills	Logistics, contracts, relationships, risk management	Market analysis, execution, risk management

بالعربية — التجارة الفعلية في السلع

تعتمد التجارة الفعلية على شراء السلع الحقيقية ونقلها وتخزينها وبيعها، ويحقق التاجر ربحه من فروقات المكان والزمان والجودة ومن الخدمات اللوجستية والتمويلية. وغالباً ما يُسعَّر العقد على أساس سعر مرجعي (مثل خام برنت أو أسعار بورصة لندن للمعادن) مضافاً إليه علاوة أو مطروحاً منه خصم. ويستخدم التاجر العقود المستقبلية للتحوط من مخاطر السعر، لكنه يبقى معرّضاً لمخاطر «الأساس»، أي الفرق بين السعر الفعلي المحلي وسعر العقد المستقبلي. والهوامش في السلع الأساسية ضيقة، لذلك فإن أي خطأ تشغيلي أو تعثر طرف مقابل قد يمحو أرباح عام كامل.

PRACTICE · Practice questions

1. A trader buys 30,000 t at futures + $40 and sells at futures + $46 a week later, hedged throughout. Freight and costs are already included in both differentials. What is the gross result?
2. Why is it dangerous for a trader to buy on a fixed price and sell on a floating price without hedging?
3. List five documents a bank might require before financing a commodity cargo.

ANSWER · Answers to the practice questions

1. Basis gain $6/t × 30,000 = **$180,000** (flat-price moves offset by the hedge).
2. If the benchmark falls between purchase and sale pricing, the floating sale price falls while the purchase cost stays fixed — an open flat-price position (speculation).
3. Purchase and sale contracts; L/C or evidence of payment security from the buyer; bill of lading (to order of the bank or endorsed); inspection certificates (quality and quantity); insurance certificate

> assigned to the bank; hedge confirmations; KYC on counterparties; collateral manager/warehouse receipts if stored.

CHAPTER 86

Energy: Crude Oil, Refined Products, Natural Gas, LNG and Coal

LEARNING OBJECTIVES · In this chapter

How energy commodities are produced, priced, shipped, traded and hedged.

86.1 Crude oil

Topic	Key points
Producers	National oil companies (e.g., Saudi Aramco, ADNOC, QatarEnergy, KPC), international oil companies, independents
Buyers	Refiners; trading houses; state buyers
Quality	API gravity (light/heavy), sulphur (sweet/sour), acidity (TAN), metals; each grade has a typical assay
Units	Barrel (bbl) = 42 US gallons ≈ 159 litres; ~7.3 bbl per tonne for medium crude (varies with density)
Benchmarks	**Brent** complex (ICE Brent futures; Dated Brent physical assessment by price reporting agencies, with WTI Midland included in the basket since 2023), **WTI** (NYMEX futures, delivered at Cushing), **Dubai/Oman** for Middle East crude to Asia, **Murban** (ICE Futures Abu Dhabi)
Pricing	Term contracts with official selling prices (OSPs) set monthly by producers as differentials to benchmarks; spot cargoes priced at benchmark averages ± differentials
Transport	Pipelines; tankers (VLCC ~2m bbl, Suezmax ~1m bbl, Aframax ~0.6–0.75m bbl)
Contracts	Producers' GT&Cs; FOB or CIF/DES deliveries; laytime/demurrage; outturn vs B/L quantity
Hedging	Brent/WTI/Dubai futures and swaps; spread hedges
Key risks	Price, freight, demurrage, quality claims, sanctions (many sanctions regimes target oil trade), counterparty

86.2 Refined products

Refineries convert crude into products: LPG, naphtha, gasoline, jet/kerosene, diesel/gasoil, fuel oil (high-sulphur and very-low-sulphur marine fuels since the IMO 2020 sulphur cap), bitumen. Prices are assessed by price reporting agencies in regional hubs (Arab Gulf, Singapore, North-West Europe, US Gulf Coast). Futures include ICE Low Sulphur Gasoil and NYMEX RBOB gasoline and ULSD. The **crack spread** (product price − crude price) measures refining margin; refiners hedge crack spreads.

WORKED EXAMPLE · Simple crack spread

Crude \$80/bbl; gasoline \$95/bbl; diesel \$102/bbl. A simplified 3-2-1 crack (3 bbl crude → 2 gasoline + 1 diesel) = (2 × 95 + 1 × 102 − 3 × 80) ÷ 3 = (190 + 102 − 240) ÷ 3 = **\$17.33/bbl** gross refining margin (before operating costs).

86.3 Natural gas and LNG

Topic	Key points
Units	MMBtu (US, Asia LNG), therm, MWh (Europe); 1 MWh ≈ 3.412 MMBtu
Hubs and benchmarks	Henry Hub (US), TTF (Netherlands, Europe's main hub), NBP (UK), JKM (LNG to Japan/Korea, assessed by a price reporting agency)

Topic	Key points
Pipeline gas	Long-term contracts, often take-or-pay, increasingly hub-indexed in Europe
LNG chain	Liquefaction (–162°C), LNG carriers (~170,000+ m^3), regasification terminals (including floating FSRUs)
LNG pricing	Long-term SPAs often oil-indexed (e.g., slope × Brent + constant) or Henry Hub-linked (Henry Hub × 115% + liquefaction fee, a common US export structure); spot cargoes at JKM or TTF-related prices
Delivery terms	FOB (buyer ships) or DES (delivered ex ship)
Major exporters	Recent years: United States, Qatar, Australia among the largest
Risks	Price, volume (take-or-pay), shipping availability, boil-off, regasification capacity, geopolitical

WORKED EXAMPLE · Oil-indexed LNG price

Contract: LNG price ($/MMBtu) = 0.13 × Brent ($/bbl) + 0.50. With Brent at $80: 0.13 × 80 + 0.5 = **$10.90/MMBtu**. A cargo of 3.4 million MMBtu is worth about **$37m**.

86.4 Coal

Thermal (steam) coal for power generation; metallurgical (coking) coal for steelmaking. Quality: calorific value (kcal/kg, on NAR or GAR basis), ash, moisture, sulphur, volatile matter. Benchmarks include API2 (CIF Amsterdam-Rotterdam-Antwerp), Newcastle (Australia FOB) and Richards Bay (South Africa FOB). Shipped in Panamax and Capesize bulk carriers. Energy-transition policies, financing restrictions by banks and insurers, and carbon pricing increasingly affect the coal trade.

PRACTICE · Practice questions

1. A 1 million bbl cargo is priced at the 5-day average of Dated Brent around the B/L date + $0.85. The average is $78.40. What is the invoice value?
2. A refiner wants to lock in its margin for next quarter. Which hedge would it use?
3. Convert 50 MWh to MMBtu.

ANSWER · Answers to the practice questions

1. (78.40 + 0.85) × 1,000,000 = **$79.25m** (adjusted for final B/L quantity).
2. Sell crack-spread swaps/futures (sell product futures and buy crude futures in the right ratios), locking the margin between crude cost and product prices.
3. 50 × 3.412 = **170.6 MMBtu**.

CHAPTER 87

Agricultural Commodities: Wheat, Corn, Sugar and Coffee

LEARNING OBJECTIVES · In this chapter

How grains and soft commodities are produced, specified, priced, shipped, financed and hedged — and why food commodities carry special political and seasonal risks.

87.1 Wheat

Topic	Key points
Producers/exporters	Russia, EU, Canada, Australia, US, Ukraine, Argentina, Kazakhstan
Major importers	Egypt, Indonesia, Turkey, Algeria, Philippines, many Middle East and North African countries
Buyers	State agencies (tenders), flour millers, feed compounders, traders
Quality	Protein %, moisture %, test weight (kg/hl), falling number, gluten, foreign matter, damaged kernels; classes (e.g., hard red winter, soft red winter, milling vs feed)
Units	Tonnes; futures in bushels (wheat bushel = 60 lb ≈ 27.22 kg) in US markets
Benchmarks	CBOT SRW wheat and KC HRW wheat (CME Group), Euronext milling wheat (Paris), assessed FOB Black Sea prices
Contracts	GAFTA standard forms; FOB or CIF/CFR; tolerance ±5–10%; quality and weight final at loading or discharge as agreed
Logistics	Panamax/Supramax/Handysize bulk carriers; port silos; discharge rates; fumigation
Risks	Weather, export restrictions (quotas, taxes, bans), war risk in the Black Sea region, sanctions screening, tender rules, quality claims, demurrage
Financing	Pre-export finance for origination; L/Cs from state buyers; inventory financing in silos

87.2 Corn (maize)

Main exporters: United States, Brazil, Argentina, Ukraine. Uses: animal feed, starch, sweeteners, ethanol. Benchmark: CBOT corn futures (corn bushel = 56 lb ≈ 25.40 kg). Quality: moisture, broken kernels, aflatoxin limits (critical for feed and food safety), GMO status (affects market access). Seasonality: northern and southern hemisphere harvests offset each other, creating seasonal export windows.

87.3 Sugar

Raw sugar (for refining) and white (refined) sugar. Leading exporters include Brazil, Thailand and India (subject to export policies). Benchmarks: **ICE Sugar No. 11** (raw sugar, cents per pound) and **ICE London No. 5** (white sugar, $/tonne); the **white premium** (white minus raw, converted to $/t) indicates refining margins. Quality: polarisation (sucrose content) for raw sugar; ICUMSA colour rating for white sugar (e.g., ICUMSA 45 for high-quality refined sugar). Shipped in bulk (raw) or bags/containers (white).

COMMON MISTAKES · Advance-fee fraud in sugar and other commodities

Unsolicited offers of large quantities of “ICUMSA 45 sugar” or other commodities at below-market prices, requiring the buyer to issue an SBLC or pay “procedure fees” before seeing goods, are a well-known fraud pattern. Legitimate trades start with verified counterparties, realistic prices and standard documentation.

87.4 Coffee

Arabica (higher quality, grown at altitude; major producers Brazil, Colombia, Ethiopia, Central America) and robusta (hardier, higher caffeine; major producers Vietnam, Brazil, Indonesia, Uganda). Benchmarks: **ICE Coffee “C”** (arabica, cents/lb) and **ICE Robusta** (London, $/t). Physical coffee trades at **differentials** to futures that depend on origin, grade, screen size, defects and cup quality; specialty coffee trades at large premiums based on cupping scores. Shipped green in containers (jute bags or bulk liners). Risks: weather and frost, quality deterioration, certification requirements, sustainability and deforestation regulations, price volatility.

بالعربية — السلع الزراعية

تتأثر السلع الزراعية بالطقس والمواسم وسياسات الدول، مثل قيود التصدير والضرائب على الصادرات. القمح يُشترى غالباً عبر مناقصات حكومية في الدول المستوردة، ويُحدَّد سعره بالرجوع إلى العقود المستقبلية أو أسعار البحر الأسود مع مواصفات دقيقة للبروتين والرطوبة. والسكر والبن يُسعَّران بفروقات عن عقود بورصة ICE. وفي كل هذه السلع تكون مواصفات الجودة وتحديد الجهة المعتمدة للفحص من أهم بنود العقد.

CHAPTER 88

Metals: Aluminium, Copper, Gold and Iron Ore

LEARNING OBJECTIVES · In this chapter

Base metals, precious metals and bulk ores: production chains, exchange benchmarks, physical premiums, contracts, financing and risks.

88.1 Base metals and the LME

The **London Metal Exchange (LME)** provides benchmark prices and futures for aluminium, copper, zinc, nickel, lead and other metals; trading also occurs on COMEX (copper) and the Shanghai Futures Exchange. LME contracts are deliverable through LME-approved warehouses via warrants. Physical metal trades at **LME price + physical premium**, where the premium reflects location (e.g., delivered to a region, duty-paid or unpaid), shape and brand, logistics and local supply-demand.

88.2 Aluminium

Produced from bauxite → alumina → smelting (very energy-intensive; Gulf countries host large smelters thanks to energy availability). Products: primary ingots (P1020 standard), billets, slabs, wire rod. Pricing: LME aluminium + regional premium (e.g., Rotterdam duty-paid/unpaid, US Midwest, Japan quarterly premium). Value-added products (billets) command additional premiums. Risks: energy costs, trade measures (tariffs, anti-dumping), premium volatility, warehouse logistics.

88.3 Copper

Mined ore → concentrate (traded with **treatment and refining charges, TC/RCs**, paid by miners to smelters) → refined cathode (Grade A, LME-deliverable) → wire rod, tubes, sheets. Major producers: Chile, Peru, DRC, China (refining). Demand linked to construction, power grids, electric vehicles and electronics. Pricing: LME/COMEX + premium; concentrates priced on payable metal content minus TC/RCs and penalties for impurities.

88.4 Gold

Topic	Key points
Units	Troy ounce = 31.1035 g; kilo bars and 400-oz London Good Delivery bars
Benchmarks	LBMA Gold Price (set twice daily by electronic auction); COMEX gold futures
Hubs	London (OTC), New York (futures), Zurich (refining), Dubai, Shanghai, Mumbai, Hong Kong
Participants	Miners, refiners, central banks, jewellery makers, investors (ETFs, bars and coins), bullion banks
Quality	Fineness (e.g., 995, 999.9); accredited refiners (LBMA Good Delivery list)
Risks	Price, fraud (counterfeit bars, false assay), money laundering, conflict-mineral sourcing — responsible sourcing rules (e.g., LBMA Responsible Gold Guidance, OECD due-diligence guidance) and AML obligations for dealers

88.5 Iron ore

Seaborne iron ore is dominated by Australia and Brazil; China is the largest importer. Products: fines, lump, pellets, concentrates. Benchmark: **62% Fe fines CFR China** indices published by price reporting agencies, with futures on SGX and the Dalian Commodity Exchange. Prices adjust for Fe content and impurities (silica, alumina, phosphorus). Shipped in Capesize and Valemax-type vessels, so freight economics (e.g., Brazil–China vs Australia–China routes) are central.

WORKED EXAMPLE · Value-in-use adjustment (iron ore, illustrative)

Index 62% Fe at \$105/dmt (dry metric tonne). A cargo of 60% Fe with Fe adjustment \$2.50 per 1% Fe and an alumina penalty of \$3/dmt: price ≈ 105 − 2 × 2.50 − 3 = **\$97/dmt**. For 170,000 wmt at 8% moisture → 156,400 dmt → invoice ≈ **\$15.17m**.

PRACTICE · Practice questions

1. A copper cathode sale: LME cash average \$9,300/t + premium \$150/t for 2,000 t. Invoice value?
2. Convert 1 kg of gold to troy ounces and value it at \$2,300/oz.
3. Why do aluminium buyers care about regional premiums as much as the LME price?

ANSWER · Answers to the practice questions

1. (9,300 + 150) × 2,000 = **\$18.9m**.
2. 1,000 ÷ 31.1035 = **32.15 oz** → ≈ **\$73,950**.
3. The premium can be a large share of the delivered price and moves independently of the LME price (driven by logistics, tariffs and regional supply); an LME hedge does not hedge the premium.

CHAPTER 89

Industrial Materials: Marble, Granite and Timber

LEARNING OBJECTIVES · In this chapter

Commodities without exchange benchmarks: natural stone and timber — specifications, pricing, logistics, inspection, regulation and risks.

89.1 Marble and granite

Topic	Key points
Producers	Quarries and processors in Italy, Turkey, Spain, Portugal, Greece, Egypt, India, China, Brazil, Iran and others
Products	Blocks (raw), slabs (cut, often 2 cm or 3 cm, polished/honed), tiles, finished products
Buyers	Fabricators, contractors, developers (hotels, towers), distributors, retailers
Specification	Stone name and origin, colour and vein pattern, grade (commercial/standard/premium/"first choice"), finish, thickness tolerance, dimensions, defects (cracks, holes, resin filling), water absorption, density, flexural strength (technical tests)
Units and pricing	Blocks per m^3 or per tonne; slabs and tiles per m^2; prices negotiated — **no exchange benchmark**; strong variation by aesthetics
Logistics	Blocks as breakbulk or in open-top/flat-rack containers; slabs in bundles on A-frames inside containers (weight limits ~20–25 t per 20ft); tiles in crates
Inspection	Buyer's visual selection at the quarry or warehouse ("block selection", photographs of each slab — "book-matched" sequences); third-party inspection for large projects
Contracts	Purchase orders with photo-approved selections, tolerance, packing, breakage allowance, payment (often advance + balance against documents)
Risks	Colour/vein variation vs approved samples, breakage in transit, project delays, buyer rejection, mis-description, weight limits on roads, project-specific quantity mismatch

PROFESSIONAL INSIGHT · Why stone trading can have higher margins

Each block and slab is unique; value comes from selection, consistency across a project, and reliable supply. Traders who understand stone, maintain quarry relationships and manage quality claims can earn margins far above bulk commodities — but working capital (stock of varied colours), breakage and claims require careful control.

89.2 Timber

Topic	Key points
Types	Softwood (pine, spruce — construction, packaging), hardwood (oak, beech, tropical species — furniture, flooring), panels (plywood, MDF, OSB), logs
Producers	Russia (subject to sanctions by several jurisdictions), Canada, US, Scandinavia, Germany, Brazil, Malaysia, New Zealand, African countries
Units	m^3 (cubic metres), board feet (US), tonnes for logs in some trades
Grading	Grading rules by species and region (knots, defects, moisture content, kiln-dried vs air-dried)
Pricing	Negotiated by grade and specification; CME lumber futures exist for North American softwood lumber

Topic	Key points
Regulation	Legality and sustainability: EU Timber Regulation and the EU Deforestation Regulation (application dates have been postponed; verify current status), US Lacey Act, certifications (FSC, PEFC); phytosanitary rules and ISPM 15 for wood packaging
Logistics	Containers (sawn timber), breakbulk (logs), moisture control, fumigation
Risks	Species misdeclaration, illegal logging, sanctions on origin, moisture damage, price swings driven by construction cycles

PRACTICE · Practice questions

1. A contractor approves marble from photographs of three slab bundles; the delivered slabs show darker veins. What contract terms should have prevented the dispute?
2. Why is timber origin due diligence important for a Gulf importer that re-exports furniture to Europe?

ANSWER · Answers to the practice questions

1. Reference to specific approved bundles/slab numbers with photographs forming part of the contract; definitions of acceptable variation; pre-shipment inspection by the buyer or its agent with authority to reject; a dispute mechanism (independent expert); breakage and rejection allowances; payment balance only after inspection or against inspection certificate.
2. EU rules on timber legality and deforestation require due diligence on the wood's origin throughout the supply chain; products from illegal or sanctioned origins can be blocked, fined or seized; the importer needs traceability documents and certifications.

CHAPTER 90

Commodity Deal Economics: Three Complete Trade P&Ls

LEARNING OBJECTIVES · In this chapter

Three fictional deals — a bulk wheat cargo, a copper cathode back-to-back trade, and a container trade of marble slabs — with full calculations of purchase price, transport, insurance, financing and other costs, selling price, gross and net margins, and the risks behind the numbers. **All figures are training assumptions.**

90.1 Deal 1 — Wheat cargo: Black Sea to North Africa

Structure: A trading company (fictional "Meridian Grains") buys 60,000 t (±5% at seller's option) of 12.5% protein milling wheat **FOB** a Black Sea port at **$230/t**, and sells to a state buyer in North Africa **CFR** at **$262/t** (won in a tender). Pricing fixed back-to-back on the same day (no flat-price exposure). Payment from the buyer: L/C at sight, confirmed by an international bank at the trader's cost. Payment to the seller: trader's L/C at sight, financed by its bank at 90% for ~35 days until the buyer's L/C pays. Quality and weight final at loading per independent inspection. Under CFR the buyer insures the cargo; the trader buys contingency (seller's interest) cover for its own exposure until it is paid (included below).

Item	Calculation	USD
Sales revenue	60,000 × 262	**15,720,000**
Purchase cost	60,000 × 230	(13,800,000)
Ocean freight (Supramax voyage charter)	60,000 × $18.00	(1,080,000)
Gross trading margin	($14.00/t)	**840,000**
Chartering broker commission	1.25% of freight	(13,500)
Contingency cargo insurance	0.08% × 110% × 15.72m	(13,834)
Inspection and superintendence (load port)	60,000 × $0.30	(18,000)
Purchase L/C issuance	0.15% × 13.8m	(20,700)
Confirmation of buyer's L/C	0.40% × 15.72m	(62,880)
Financing	90% × 13.8m × 7.5% × 35/360	(90,563)
Demurrage (net of despatch)	estimated $0.50/t	(30,000)
Quality/weight claim reserve	0.2% of sales	(31,440)
Net trading profit (before overheads and tax)	**$9.32/t; 3.6% of sales**	**559,083**

Sensitivities and failure scenarios:

Event	Impact
Freight rises $3/t before vessel fixed (fixed after the sale)	−$180,000 (hedge with freight forward agreements or fix vessel early)
Discharge delayed 6 days beyond laytime at $22,000/day demurrage, not recoverable from buyer	−$132,000
Documentary discrepancy causing 20-day payment delay	Extra financing ≈ −$52,000

Event	Impact
Buyer refuses cargo (with confirmed L/C, compliant documents are paid by the confirming bank)	Protected if documents comply
Seller defaults (does not load) after prices rose $15/t	Trader must buy replacement cargo ≈ –$900,000, then claim against seller under GAFTA default clause — recovery uncertain
Export restriction imposed by the origin country	Force majeure/prohibition clauses determine outcome; potential large loss

Lesson: a $14/t gross margin becomes $9/t after costs, and several plausible events can erase it. Disciplined chartering, documentation, counterparty selection and contract terms are what make the business profitable.

90.2 Deal 2 — Copper cathodes: back-to-back with a pricing-period mismatch

Structure: Meridian Metals (fictional) buys 500 t of LME Grade A copper cathodes **CIF Gemlik (Turkey)** from a South American producer at **LME cash average of month M + $110/t**, paying by sight L/C. It sells to a Turkish wire-rod maker **delivered to the buyer's plant** at **LME cash average of month M+1 + $285/t**, payment 30 days after delivery (credit-insured). The mismatch of pricing periods creates price exposure, hedged on the LME: the trader **sells** futures as the purchase price is fixed during month M and **buys them back** as the sale price is fixed during month M+1.

Outcome: LME average month M = $9,400; month M+1 = $9,550.

Item	Calculation	USD
Sales revenue	500 × (9,550 + 285)	4,917,500
Purchase cost	500 × (9,400 + 110)	(4,755,000)
Physical result		**162,500**
LME hedge result	Sold at avg 9,400, bought back at avg 9,550: –150 × 500	(75,000)
Hedged gross margin (= premium spread 285 – 110 = $175/t)		**87,500**
Port handling, customs formalities, trucking to plant	500 × $18	(9,000)
Inland transit and stock insurance	0.03% of sales	(1,475)
L/C costs (purchase)	0.15%	(7,133)
Financing (purchase price outstanding ~45 days until buyer pays)	4.755m × 6.5% × 45/360	(38,634)
Credit insurance on buyer	0.25% of sales	(12,294)
LME execution and clearing fees	20 lots each way	(400)
Net profit	**$37/t; 0.38% of sales**	**18,564**

Lesson: without the hedge the trader would have made $93,564 — or, if copper had fallen $150, lost $56,436. The hedge converts a speculative result into a predictable premium margin. Most of the gross margin pays for **credit and financing** provided to the buyer; the net margin is very thin, so volume, financing costs and operational precision drive profitability.

90.3 Deal 3 — Marble slabs from Turkey to Kuwait

Structure: Levant Stone Trading (fictional) buys 3,600 m² of 2 cm polished white marble slabs (photo-approved bundles) from a Turkish processor at **$30/m² FOB Izmir**: 30% advance, 70% before loading. It sells to a Kuwaiti contractor for a hotel project at **$46/m² CFR Shuwaikh**: 30% advance, 70% by documentary collection (D/P). Shipment: 8 × 20ft containers (~450 m² each).

Item	Calculation	USD
Sales revenue	3,600 × 46	**165,600**
Purchase cost	3,600 × 30	(108,000)
Ocean freight Izmir → Kuwait	8 × $1,900	(15,200)
Gross margin	(25.6%)	**42,400**
Cargo insurance incl. breakage clause	0.3% × 110% × 123,200	(407)
Quality agent at processor (selection, loading supervision)		(1,500)
Breakage allowance (credit note to buyer)	3% of sales	(4,968)
Colour-variation claim reserve	2% of sales	(3,312)
Bank collection charges	0.2% of sales	(331)
Financing (advances and inventory before buyer pays)	~$75,600 for ~60 days at 8% (approx.)	(1,600)
Net profit	**18.3% of sales**	**30,282**

Risks: buyer refuses documents under D/P (goods stranded in Kuwait with demurrage); colour mismatch claims larger than reserved; project delays leading the contractor to postpone delivery; breakage above allowance; processor substitutes lower-grade slabs. Controls: photo-numbered slab lists in both contracts, loading supervision, higher advance from the buyer or L/C for new customers, insurance with breakage cover, and clear claim procedures.

90.4 Comparing the three deals

	Wheat (bulk)	Copper (exchange metal)	Marble (speciality)
Deal value	$15.7m	$4.9m	$0.17m
Net margin	3.6%	0.4%	18.3%
Net profit	$559k	$19k	$30k
Main value driver	Freight and logistics, tender pricing	Premium spread and credit service	Selection and quality management
Main risks	Performance, freight, demurrage, country	Basis/premium, credit, financing cost	Quality claims, buyer acceptance
Capital intensity	High (financed by banks)	High	Low

بالعربية — ربحية الصفقات في السلع

توضح الصفقات الثلاث أن نسبة الهامش وحدها لا تكفي للحكم على الصفقة: صفقة القمح تحقق هامشاً صغيراً كنسبة لكنها كبيرة القيمة، وتعتمد على إدارة الشحن والمستندات والطرف المقابل. صفقة النحاس هامشها ضئيل جداً لأن معظم الربح الإجمالي يذهب لتغطية التمويل والائتمان الممنوح للمشتري، ولولا التحوط لكانت النتيجة مضاربة. أما صفقة الرخام فهامشها مرتفع لكن مخاطر الجودة ورفض البضاعة كبيرة.

PRACTICE · Practice questions

1. In Deal 1, recompute the net profit if the buyer's L/C confirmation costs 0.75% instead of 0.40%, and demurrage is $2.00/t instead of $0.50/t.
2. In Deal 2, what would the net profit have been without the hedge if the LME average for month M+1 had been $9,250?
3. In Deal 3, the contractor asks for 90-day open account instead of D/P. What would you require?

ANSWER · Answers to the practice questions

1. Extra confirmation: 0.35% × 15.72m = $55,020; extra demurrage: $1.50 × 60,000 = $90,000 → net ≈ 559,083 − 55,020 − 90,000 = **$414,063** ($6.90/t).
2. Physical result: 500 × (9,250 + 285) − 4,755,000 = 4,767,500 − 4,755,000 = 12,500; less costs ex-hedge (68,936) → **−$56,436** (a loss) without the hedge; with the hedge the result would still be +**$18,564** (hedge gain +75,000).
3. Credit assessment of the contractor; credit insurance or a bank guarantee/SBLC; higher advance; price increase for the 90-day credit (≈ 2% at 8% p.a.); retention of title clause (subject to Kuwaiti law); and monitoring of project payment status.

SUMMARY · Part XI summary

- Physical traders earn from location, time, quality, logistics and credit services — not from guessing prices.
- Contracts use benchmarks ± differentials; hedging removes flat-price risk but leaves basis and premium risk.
- Quality, quantity, inspection and delivery terms are central to value and disputes.
- Bulk margins are thin and erased by small operational failures; speciality trades earn more but carry quality risk.
- Physical trading is a logistics, credit and risk-management business with heavy financing needs.

PART XXI

How to Read Business News and Reports

كيف تقرأ الأخبار والتقارير الاقتصادية

Professionals are constantly fed information: earnings releases, central-bank statements, analyst notes, rating actions, commodity reports, shipping indices, government announcements and news headlines. The skill is not reading more but reading better: knowing which numbers matter, separating facts from estimates and spin, asking the right questions, and connecting information to decisions. This part provides reading guides, annotated (fictional) examples and checklists.

LEARNING OBJECTIVES · **Learning objectives for Part XXI**

- Read annual and quarterly reports, earnings releases, press releases and merger announcements efficiently.
- Interpret bank financial statements and the key indicators of bank health.
- Understand central-bank decisions, inflation reports, economic forecasts and policy announcements.
- Use investment research, credit-rating reports, industry, commodity and shipping reports critically.
- Distinguish facts, estimates, management claims, market expectations and speculation.
- Translate information into business implications and actions.

Chapters in this part

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Chapter 92 Reading Macroeconomic Information: Central Banks, Rate Decisions, Inflation Reports, Forecasts and Government Policy 48

Chapter 93 Reading Research, Ratings, Industry, Commodity and Shipping Reports — and Separating Facts from Opinions 50

CHAPTER 91

Reading Company Reports: Annual Reports, Quarterly Earnings, Press Releases and Merger Announcements

LEARNING OBJECTIVES · In this chapter

What to read first, which numbers matter, how to detect spin, and how to read bank statements and M&A announcements.

91.1 The annual report: a reading order

Order	Section	What to look for
1	Auditor's report	Opinion type, going-concern section, key audit matters
2	Financial highlights / KPIs	Revenue, margins, cash flow, leverage — note which metrics management chose
3	Cash flow statement	Operating cash vs profit, capex, acquisitions, dividends, debt flows
4	Balance sheet	Debt, liquidity, working capital, goodwill, provisions
5	Income statement	Revenue growth drivers, margins, one-offs, effective tax rate
6	Notes	Segments, debt maturities and covenants, related parties, contingencies, leases, subsequent events, accounting policy changes
7	Management commentary (MD&A / strategic report)	Strategy, outlook, risk factors — compare with numbers
8	Governance report	Board composition, remuneration, related-party oversight
9	Chairman's and CEO's letters	Tone and promises — read last, critically

91.2 Quarterly earnings releases

Key items: revenue and EPS vs **consensus expectations** (analysts' average forecast), margins, guidance changes, segment performance, cash flow, net debt, one-off items, and management's explanation. Market reaction depends on **results versus expectations** and **guidance**, not on absolute results: a company can report record profits and see its share price fall because the market expected more.

TRAINING SIMULATION · FICTIONAL DOCUMENT · NOT A VALID INSTRUMENT

Earnings release (fictional, annotated)

Gulf Packaging Industries (fictional) — Q3 results "Record quarterly revenue of $212m (+14% y/y) and adjusted EBITDA of $41m (+11%). Net income of $18m. We are raising full-year revenue guidance to $800–820m. Free cash flow was $(6)m due to temporary working-capital investment."

Annotations:

- *"Record revenue"*: +14% — how much from price increases (resin pass-through), volume, acquisitions, FX? Organic volume growth may be small.
- *"Adjusted EBITDA +11%"* grew slower than revenue → margin fell (from ~19.9% to ~19.3%). What was adjusted out?
- *Net income $18m*: compare with last year and consensus; check one-offs.

- *Guidance raised on revenue only*: no profit guidance raise — margins under pressure?
- *Negative FCF "temporary"*: check receivables and inventory growth; is this the third "temporary" quarter?
- *Missing*: net debt, covenant headroom, order backlog.

91.3 Spotting spin

Phrase	Possible meaning
"Adjusted", "underlying", "core", "pro forma"	Excludes items management prefers you ignore — check reconciliation to IFRS figures
"Temporary working-capital investment"	Possible collection problems or stock build-up
"Challenging market conditions"	Results missed plan; is it the market or the company (compare peers)?
"Strategic review"	Possibly a sale, closure or restructuring
"One-off restructuring costs" (every year)	Recurring costs presented as exceptional
"Record"	True but maybe meaningless (inflation, acquisitions)
"Robust pipeline"	Not yet revenue — what conversion rate?
"Remain confident"	Often precedes a guidance cut

91.4 Reading bank financial statements

Area	Indicators	What they reveal
Profitability	NIM, cost-to-income, ROE, fee income share	Business model strength
Asset quality	NPL ratio, Stage 2 and 3 loans (IFRS 9), coverage ratio, cost of risk, restructured loans, sector concentrations	Credit risk
Capital	CET1, Tier 1, total capital ratios vs requirements and buffers; leverage ratio	Loss-absorbing capacity
Liquidity and funding	LCR, NSFR, loan-to-deposit ratio, deposit mix (current vs term), wholesale funding reliance	Resilience to outflows
Market risk	Securities portfolio, unrealised losses (fair value vs amortised cost), interest-rate sensitivity	Rate shocks (e.g., 2023 US regional bank stress)
Concentration	Top 20 borrowers, related-party lending, government exposure	Fragility

PROFESSIONAL INSIGHT · Why this matters even if you never work in a bank

Your company's deposits, guarantees and credit lines depend on your banks' health. A corporate treasurer reads bank results to set counterparty limits, and an exporter relying on a foreign issuing bank's L/C should know that bank's capital and asset quality — or obtain confirmation.

91.5 Merger and acquisition announcements

Read for: **price** (premium to undisturbed share price; EV/EBITDA multiple), **consideration** (cash, shares, mix), **financing** (new debt, equity), **strategic rationale**, **synergies** (amount, timing, cost to achieve), **accretion/dilution claims**, **conditions** (regulatory approvals, shareholder votes), **timeline**, **break fees**, and **management/governance** after the deal. Compare synergies with the premium paid (Chapter 35 (Volume 3)).

PRACTICE · Practice questions

1. A company reports net income up 25% while operating cash flow fell 40%. List the first three notes you would read.
2. A release states “EPS accretive in year one”. What else do you need to judge the deal?
3. Which bank indicators would you check before placing $20m of deposits with a regional bank?

ANSWER · Answers to the practice questions

1. Trade receivables (ageing, allowance), inventories (composition, provisions), revenue recognition policy and contract assets; also other income/gains.
2. Price paid vs intrinsic value, synergies vs premium, financing (leverage, cost of debt), ROIC vs WACC, integration risks, one-off costs excluded from “accretion”.
3. Credit rating and outlook, CET1 and total capital ratios with buffers, NPL ratio and coverage, LCR/NSFR, deposit concentration, profitability trend, and any unrealised securities losses — plus diversification of your own deposits across banks.

CHAPTER 92

Reading Macroeconomic Information: Central Banks, Rate Decisions, Inflation Reports, Forecasts and Government Policy

LEARNING OBJECTIVES · In this chapter

How to read central-bank statements, rate decisions, inflation data, economic forecasts and government policy announcements, and how to translate them into business implications.

92.1 Central-bank decisions and statements

Read: the **decision** (rate change and size), **vote split**, **statement language changes** versus the previous statement, **forward guidance**, **projections** (inflation, growth, policy-rate path where published), and the **press conference** tone. Markets react to surprises relative to expectations (priced in futures and swaps).

TRAINING SIMULATION · FICTIONAL DOCUMENT · NOT A VALID INSTRUMENT

Central-bank statement (fictional, annotated)

"The Monetary Policy Committee decided to keep the policy rate unchanged at 5.25%. Inflation has declined but remains above target, and services inflation is persistent. The Committee **will remain attentive** to upside risks and **is prepared to adjust policy as needed**. Two members voted for a 25 basis point reduction."

Annotations: unchanged rate (expected?), but two votes for a cut — a **dovish** signal that cuts may come soon; "persistent services inflation" keeps some caution. Business implications: floating-rate borrowers may see lower costs in coming months; consider timing of fixed-rate borrowing; local currency may weaken slightly if markets bring forward rate cuts.

Term	Meaning
Hawkish	Leaning toward tighter policy (higher rates) to fight inflation
Dovish	Leaning toward looser policy (lower rates) to support growth
Data-dependent	Decisions will follow incoming data — no pre-commitment
Neutral rate	Rate that neither stimulates nor restrains the economy
Pause / hold	No change
Pivot	Change in direction of policy
Quantitative tightening	Shrinking the central-bank balance sheet

92.2 Inflation reports

Look at headline vs core inflation, month-on-month vs year-on-year rates, components (food, energy, housing, services, goods), producer prices (pipeline pressures), wage growth and inflation expectations. Base effects (last year's comparison) can distort year-on-year rates.

92.3 Economic forecasts

Forecasts from the IMF (World Economic Outlook), World Bank, OECD, central banks, ministries and private economists differ in assumptions. Read the **assumptions** (oil price, interest rates, fiscal policy), the **range** of scenarios and the **track record**. Use forecasts as inputs for scenarios, not as predictions.

92.4 Government policy announcements

Budgets (spending priorities, deficits, project pipelines), tax changes (VAT, corporate tax, excise), trade measures (tariffs, import bans, local-content rules), labour and localisation rules, subsidies and price controls, investment incentives, licensing and regulatory reforms. Ask: when does it take effect, who is affected, what transition rules apply, and what implementing regulations are still to come?

92.5 From macro news to business action

News	Questions to ask	Possible actions
Rate cut expected	How much floating debt? Customer demand sensitivity?	Delay fixing rates; prepare for demand pick-up
Inflation surprise up	Cost pass-through ability? Fixed-price contracts?	Accelerate price reviews; indexation clauses
Currency weakening	Import share of costs? FX debt?	Hedge payables; reprice; local sourcing
Government capex cut	Revenue from public projects?	Diversify customers; tighten credit to contractors
New tariff on imports	Sourcing origin options? Competitors affected?	Re-source; adjust prices; use FTAs
Oil price fall (oil exporter economy)	Government payment delays? Consumer demand?	Monitor receivables; adjust budgets

PRACTICE · Practice questions

1. Headline inflation fell from 4.5% to 3.2%, while core inflation rose from 3.8% to 4.0%. What might the central bank conclude?
2. A government announces a new 15% tariff on imported tiles effective in 6 months. You import tiles. List four actions.

ANSWER · Answers to the practice questions

1. The fall is likely driven by volatile components (energy/food); underlying inflation is still rising, so the central bank may stay cautious (hawkish) rather than cut rates.
2. Bring forward imports before the effective date (if storage and cash allow); seek suppliers from FTA-partner countries with qualifying origin; renegotiate supplier prices; review selling prices and contracts with customers (pass-through clauses); assess domestic suppliers; check transitional rules for goods in transit.

CHAPTER 93

Reading Research, Ratings, Industry, Commodity and Shipping Reports — and Separating Facts from Opinions

LEARNING OBJECTIVES · In this chapter

How to use sell-side and independent research, credit-rating reports, industry studies, commodity market reports and shipping data critically — and a framework to classify every statement you read.

93.1 Investment research

Sell-side research (from brokers and investment banks) provides models, forecasts, ratings (buy/hold/sell) and price targets. Use it for information and models, but recognise potential conflicts of interest (investment-banking relationships, access to management), herding, and optimism bias. Read the **assumptions**, **valuation method**, **risks** section and the analyst's **track record**. Independent research and your own analysis remain essential.

93.2 Credit-rating reports

A rating report explains the rating, outlook (positive/stable/negative) or watch status, key rating drivers, sensitivities ("what could lead to an upgrade/downgrade"), financial metrics and peer comparisons. The **sensitivities** section is the most useful: it tells you the thresholds (e.g., "net leverage above 3.5× for a sustained period") that would trigger rating changes — useful for lenders, bond investors and the company's own financial policy.

TRAINING SIMULATION · FICTIONAL DOCUMENT · NOT A VALID INSTRUMENT

Rating action (fictional, annotated)

"The agency has affirmed Levant Telecom's (fictional) long-term rating at BB with a Negative outlook. The outlook reflects rising leverage following debt-funded spectrum payments and FX pressure on USD-denominated debt. **The rating could be downgraded if** funds from operations net leverage exceeds 4.0× or liquidity falls below 12 months of debt service. **Positive action is unlikely** before leverage falls below 3.0×."

Annotations: affirmed (no change now), but Negative outlook means a meaningful chance of downgrade within 12–24 months; watch FX and leverage; bond prices may already reflect risk; suppliers and lenders should monitor liquidity.

93.3 Industry reports

From consultancies, industry associations, government statistics and specialist publishers. Check: who paid for it, methodology, definitions (market size of what, exactly?), base year, forecast assumptions (often optimistic CAGRs), and consistency with other sources. Use bottom-up checks for any market-size number.

93.4 Commodity market reports

Examples of widely used sources: monthly oil market reports from the IEA and OPEC, the US Energy Information Administration's data and outlooks, the USDA's World Agricultural Supply and Demand Estimates (WASDE) for grains and oilseeds, the International Grains Council, metals statistics from industry

groups, and price assessments from price reporting agencies. Read: supply, demand and inventory balances; revisions from previous reports; stocks-to-use ratios (for grains); OECD commercial stocks and forward curves (for oil); and the reasons for changes.

93.5 Shipping and trade reports

Indicator	What it measures
Baltic Dry Index (BDI) and sub-indices (Capesize, Panamax, Supramax)	Dry bulk freight rates on key routes
Container freight indices (e.g., Shanghai Containerized Freight Index; Drewry World Container Index; Freightos Baltic Index)	Spot container rates on major lanes
Tanker rates (Worldscale-based assessments, time-charter equivalents)	Oil shipping costs
Port congestion and vessel waiting times	Logistics delays, D&D risk
Schedule reliability data	Carrier punctuality
Trade statistics (customs, UN Comtrade, national statistics)	Volumes by product and origin

Use them to time freight contracts, anticipate cost changes, and understand demand trends (e.g., falling container rates may signal weakening import demand or overcapacity).

93.6 Facts, estimates, claims, expectations and speculation

Category	Definition	Example	How to treat it
Fact	Verifiable, already happened, ideally audited or officially published	"Q3 revenue was $212m (reported, unaudited)"	Verify source and definitions
Estimate	Quantitative judgement about the present or past where data are incomplete	"The market is worth about $1.2bn"	Ask for method and range
Management claim	Statement by an interested party	"Best-in-class margins", "demand remains strong"	Test against data and peers
Market expectation	What prices and consensus imply	"Futures price two rate cuts next year"; "consensus EPS $2.10"	Useful baseline; surprises move prices
Forecast	Forward-looking projection with assumptions	"Revenue to grow 15% a year to 2030"	Check assumptions, track record, scenarios
Speculation / rumour	Unconfirmed, often anonymous	"Sources say the company is exploring a sale"	Do not act on it alone; wait for confirmation; beware market-abuse rules
Opinion	Judgement or view	"The stock is undervalued"	Consider the author's incentives

بالعربية — التمييز بين الحقائق والتقديرات والآراء

عند قراءة أي تقرير أو خبر اقتصادي اسأل: هل هذه حقيقة منشورة رسمياً أو مدققة؟ أم تقدير يحتاج إلى معرفة المنهجية؟ أم ادعاء من الإدارة يجب اختباره بالأرقام ومقارنته بالمنافسين؟ أم توقعات السوق المتضمنة في الأسعار؟ أم مجرد إشاعة؟ ومن المستفيد من هذه الرسالة؟ هذا التمييز يحميك من اتخاذ قرارات مبنية على معلومات غير موثوقة، ويجعل أسئلتك في الاجتماعات أكثر دقة.

93.7 A reading checklist

1. **Source:** who published it, with what incentives and methodology?

2. **Date:** current? which period does it cover?
3. **Definitions:** what exactly is measured (gross vs net, adjusted vs reported, nominal vs real)?
4. **Comparison:** versus last period, last year, expectations, peers, targets.
5. **Drivers:** why did it change?
6. **Classification:** fact, estimate, claim, expectation, forecast, speculation?
7. **Implications:** for our revenue, costs, cash, risks, counterparties, decisions.
8. **Action:** what will we do, monitor, or ask?

PRACTICE · Practice questions

1. Classify: (a) “Brent averaged $82 in Q2” (price reporting agency data); (b) “We expect to double market share in three years” (CEO); (c) “Analysts forecast EPS of $3.40”; (d) “Traders say a large producer may cut output”.
2. Container spot rates on Asia–Gulf lanes fell 35% in two months. What could this mean for an importer and how might it respond?
3. A rating report says a downgrade could follow if “EBITDA interest cover falls below 2.5×”. You are a supplier offering 90-day credit. What do you do with this information?

ANSWER · Answers to the practice questions

1. (a) Fact (published market data); (b) management claim/forecast; (c) market expectation (consensus forecast); (d) speculation/rumour.
2. Possibly weaker demand or overcapacity; for the importer, an opportunity to renegotiate freight contracts, shift from long-term contracts to shorter ones or index-linked rates, review landed costs and pricing — but check whether surcharges or capacity cuts (blank sailings) may reverse rates.
3. Monitor the customer’s interest cover in its published reports; set your credit limit and terms with that threshold in mind (e.g., reduce exposure or require security if cover approaches 2.5×); consider credit insurance.

SUMMARY · Part XXI summary

- Read audits, cash flows and notes before management narratives.
- Results matter relative to expectations and guidance; adjusted metrics need reconciliation.
- Central-bank language changes and votes signal future policy; translate macro news into concrete business actions.
- Use research, ratings, industry and commodity reports critically: sources, assumptions and incentives.
- Always classify information: fact, estimate, claim, expectation, forecast or speculation.

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Bold page numbers mark the main term-card definition.